DBMS MINI PROJECT

PROJECT TITLE

A SYSTEM OF SMART POWER GRID MANAGEMENT AND ENERGY RESOURCE ALLOCATION

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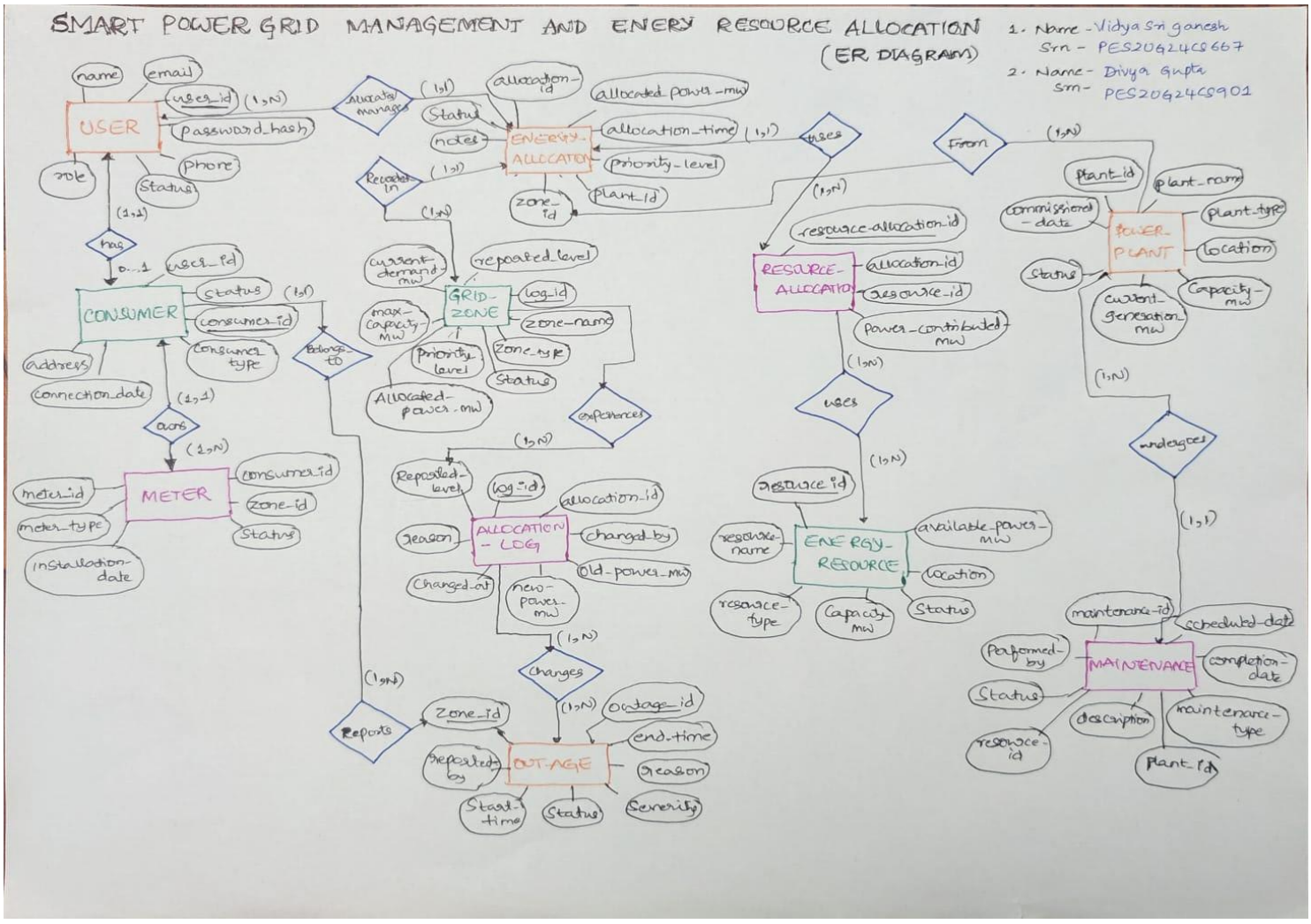
SRN -PES2UG24CS901

Team Mem2- Vidya Sri Ganesh

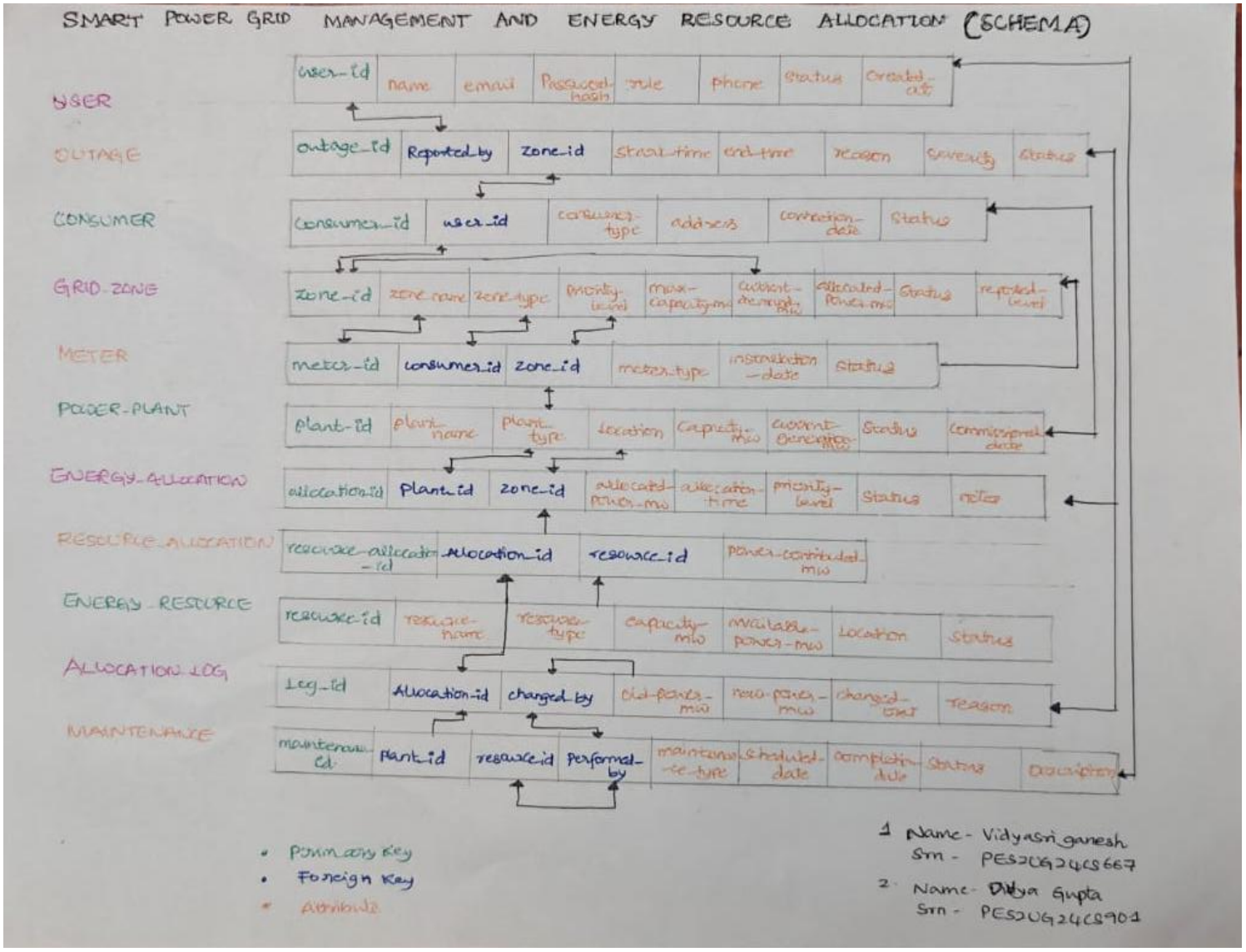
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Section 5K

1. ER DIAGRAM



2. RELATIONAL SCHEMA



3. SQL Implementation – DDL Statements

Step 0 – Create Database

SCREENSHOT 1

```
mysql> CREATE DATABASE smart_power_grid;
Query OK, 1 row affected (0.10 sec)

mysql> USE smart_power_grid;
Database changed
```

Step 1 – USER

SCREENSHOT 2

```
mysql> CREATE TABLE USER (
    ->     user_id INT PRIMARY KEY AUTO_INCREMENT,
    ->     name VARCHAR(100) NOT NULL,
    ->     email VARCHAR(150) NOT NULL UNIQUE,
    ->     password_hash VARCHAR(255) NOT NULL,
    ->     role ENUM('ADMIN', 'OPERATOR', 'CONSUMER') NOT NULL,
    ->     phone VARCHAR(20),
    ->     status ENUM('ACTIVE', 'INACTIVE') NOT NULL DEFAULT 'ACTIVE',
    ->     created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
    -> );
Query OK, 0 rows affected (0.22 sec)
```

Step 2 – GRID_ZONE

SCREENSHOT 3

```
mysql> CREATE TABLE GRID_ZONE (
    ->     zone_id INT PRIMARY KEY AUTO_INCREMENT,
    ->     zone_name VARCHAR(100) NOT NULL,
    ->     zone_type ENUM('RESIDENTIAL', 'COMMERCIAL', 'INDUSTRIAL', 'CRITICAL')
 NOT NULL,
    ->     priority_level INT NOT NULL,
    ->     max_capacity_mw DECIMAL(10,2) NOT NULL,
    ->     current_demand_mw DECIMAL(10,2) NOT NULL DEFAULT 0,
    ->     allocated_power_mw DECIMAL(10,2) NOT NULL DEFAULT 0,
    ->     status ENUM('NORMAL', 'HIGH_DEMAND', 'OUTAGE') NOT NULL DEFAULT 'NORM
AL',
    ->     reported_level VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.12 sec)
```

Step 3 – POWER_PLANT

SCREENSHOT 4

```
mysql> CREATE TABLE POWER_PLANT (
    ->     plant_id INT PRIMARY KEY AUTO_INCREMENT,
    ->     plant_name VARCHAR(150) NOT NULL,
    ->     plant_type ENUM('SOLAR', 'WIND', 'HYDRO', 'THERMAL') NOT NULL,
    ->     location VARCHAR(200) NOT NULL,
    ->     capacity_mw DECIMAL(10,2) NOT NULL,
    ->     current_generation_mw DECIMAL(10,2) NOT NULL DEFAULT 0,
    ->     status ENUM('ACTIVE', 'OFFLINE', 'MAINTENANCE') NOT NULL DEFAULT 'ACT
IVE',
    ->     commissioned_date DATE
    -> );
Query OK, 0 rows affected (0.11 sec)
```

Step 4 – CONSUMER

SCREENSHOT 5

```
mysql> CREATE TABLE CONSUMER (
    ->     consumer_id INT PRIMARY KEY AUTO_INCREMENT,
    ->     user_id INT NOT NULL,
    ->     consumer_type ENUM('RESIDENTIAL', 'COMMERCIAL', 'INDUSTRIAL') NOT NUL
L,
    ->     address VARCHAR(255) NOT NULL,
    ->     connection_date DATE NOT NULL,
    ->     status ENUM('ACTIVE', 'DISCONNECTED') NOT NULL DEFAULT 'ACTIVE',
    ->
    ->     CONSTRAINT fk_consumer_user
    ->         FOREIGN KEY (user_id)
    ->         REFERENCES USER(user_id)
    -> );
Query OK, 0 rows affected (0.17 sec)
```

Step 5 – OUTAGE

SCREENSHOT 6

```
mysql> CREATE TABLE OUTAGE (
    ->     outage_id INT PRIMARY KEY AUTO_INCREMENT,
    ->     zone_id INT NOT NULL,
    ->     reported_by INT NOT NULL,
    ->     start_time DATETIME NOT NULL,
    ->     end_time DATETIME,
    ->     reason VARCHAR(255),
    ->     severity ENUM('LOW', 'MEDIUM', 'HIGH', 'CRITICAL') NOT NULL,
    ->     status ENUM('ACTIVE', 'RESOLVED') NOT NULL DEFAULT 'ACTIVE',
    ->
    ->     CONSTRAINT fk_outage_zone
    ->         FOREIGN KEY (zone_id)
    ->         REFERENCES GRID_ZONE(zone_id),
    ->
    ->     CONSTRAINT fk_outage_user
    ->         FOREIGN KEY (reported_by)
    ->         REFERENCES USER(user_id),
    ->
    ->     CONSTRAINT chk_outage_time
    ->         CHECK (end_time IS NULL OR end_time >= start_time)
    -> );
Query OK, 0 rows affected (0.11 sec)
```

Step 6 – METER

SCREENSHOT 7

```
mysql> CREATE TABLE METER (
    ->     meter_id INT PRIMARY KEY AUTO_INCREMENT,
    ->     consumer_id INT NOT NULL,
    ->     zone_id INT NOT NULL,
    ->     meter_type ENUM('SMART', 'STANDARD') NOT NULL,
    ->     installation_date DATE NOT NULL,
    ->     status ENUM('ACTIVE', 'FAULTY', 'INACTIVE') NOT NULL DEFAULT 'ACTIVE'
'
    ->
    ->     CONSTRAINT fk_meter_consumer
    ->         FOREIGN KEY (consumer_id)
    ->         REFERENCES CONSUMER(consumer_id),
    ->
    ->     CONSTRAINT fk_meter_zone
    ->         FOREIGN KEY (zone_id)
    ->         REFERENCES GRID_ZONE(zone_id)
    -> );
Query OK, 0 rows affected (0.08 sec)
```

Step 7 – ENERGY_ALLOCATION

SCREENSHOT 8

```
mysql> CREATE TABLE ENERGY_ALLOCATION (
    ->     allocation_id INT PRIMARY KEY AUTO_INCREMENT,
    ->     plant_id INT NOT NULL,
    ->     zone_id INT NOT NULL,
    ->     allocated_power_mw DECIMAL(10,2) NOT NULL,
    ->     allocation_time DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
    ->     priority_level INT NOT NULL,
    ->     status ENUM('PENDING', 'APPROVED', 'COMPLETED') NOT NULL DEFAULT 'PEN
DING',
    ->     notes VARCHAR(500),
    ->
    ->     CONSTRAINT fk_allocation_plant
    ->         FOREIGN KEY (plant_id)
    ->         REFERENCES POWER_PLANT(plant_id),
    ->
    ->     CONSTRAINT fk_allocation_zone
    ->         FOREIGN KEY (zone_id)
    ->         REFERENCES GRID_ZONE(zone_id),
    ->
    ->     CONSTRAINT chk_allocation_power
    ->         CHECK (allocated_power_mw > 0)
    -> );
Query OK, 0 rows affected (0.12 sec)
```

Step 8 – ENERGY_RESOURCE

SCREENSHOT 9

```
mysql> CREATE TABLE ENERGY_RESOURCE (
    ->     resource_id INT PRIMARY KEY AUTO_INCREMENT,
    ->     resource_name VARCHAR(150) NOT NULL,
    ->     resource_type ENUM('BATTERY', 'GENERATOR', 'RESERVE', 'STORAGE') NOT
NULL,
    ->     capacity_mw DECIMAL(10,2) NOT NULL,
    ->     available_power_mw DECIMAL(10,2) NOT NULL DEFAULT 0,
    ->     location VARCHAR(200),
    ->     status ENUM('AVAILABLE', 'IN_USE', 'MAINTENANCE') NOT NULL DEFAULT 'A
VAILABLE',
    ->
    ->     CONSTRAINT chk_resource_capacity
    ->         CHECK (capacity_mw >= 0),
    ->
    ->     CONSTRAINT chk_resource_available
    ->         CHECK (available_power_mw >= 0 AND available_power_mw <= capacity
_mw)
    -> );
Query OK, 0 rows affected (0.10 sec)
```

Step 9 – RESOURCE_ALLOCATION

SCREENSHOT 10

```
mysql> CREATE TABLE RESOURCE_ALLOCATION (
    ->     resource_allocation_id INT PRIMARY KEY AUTO_INCREMENT,
    ->     allocation_id INT NOT NULL,
    ->     resource_id INT NOT NULL,
    ->     power_contributed_mw DECIMAL(10,2) NOT NULL,
    ->
    ->     CONSTRAINT fk_resource_allocation
    ->         FOREIGN KEY (allocation_id)
    ->         REFERENCES ENERGY_ALLOCATION(allocation_id),
    ->
    ->     CONSTRAINT fk_resource
    ->         FOREIGN KEY (resource_id)
    ->         REFERENCES ENERGY_RESOURCE(resource_id),
    ->
    ->     CONSTRAINT chk_power_contributed
    ->         CHECK (power_contributed_mw > 0)
    -> );
Query OK, 0 rows affected (0.10 sec)
```

Step 10 – ALLOCATION_LOG

SCREENSHOT 11

```
mysql> CREATE TABLE ALLOCATION_LOG (
    ->     log_id INT PRIMARY KEY AUTO_INCREMENT,
    ->     allocation_id INT NOT NULL,
    ->     changed_by INT NOT NULL,
    ->     old_power_mw DECIMAL(10,2),
    ->     new_power_mw DECIMAL(10,2),
    ->     changed_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
    ->     reason VARCHAR(255),
    ->
    ->     CONSTRAINT fk_log_allocation
    ->         FOREIGN KEY (allocation_id)
    ->         REFERENCES ENERGY_ALLOCATION(allocation_id),
    ->
    ->     CONSTRAINT fk_log_user
    ->         FOREIGN KEY (changed_by)
    ->         REFERENCES USER(user_id)
    -> );
Query OK, 0 rows affected (0.11 sec)
```

Step 11 – MAINTENANCE

SCREENSHOT 12

```
mysql> CREATE TABLE MAINTENANCE (
    ->     maintenance_id INT PRIMARY KEY AUTO_INCREMENT,
    ->     plant_id INT,
    ->     resource_id INT,
    ->     maintenance_type ENUM('ROUTINE', 'REPAIR', 'EMERGENCY') NOT NULL,
    ->     scheduled_date DATE NOT NULL,
    ->     completion_date DATE,
    ->     status ENUM('SCHEDULED', 'IN_PROGRESS', 'COMPLETED') NOT NULL DEFAULT
 'SCHEDULED',
    ->     performed_by INT NOT NULL,
    ->     description VARCHAR(500),
    ->
    ->     CONSTRAINT fk_maintenance_plant
    ->         FOREIGN KEY (plant_id)
    ->         REFERENCES POWER_PLANT(plant_id),
    ->
    ->     CONSTRAINT fk_maintenance_resource
    ->         FOREIGN KEY (resource_id)
    ->         REFERENCES ENERGY_RESOURCE(resource_id),
    ->
    ->     CONSTRAINT fk_maintenance_user
    ->         FOREIGN KEY (performed_by)
    ->         REFERENCES USER(user_id),
    ->
    ->     CONSTRAINT chk_maintenance_target
    ->         CHECK (
    ->             (plant_id IS NOT NULL AND resource_id IS NULL)
    ->             OR
    ->             (plant_id IS NULL AND resource_id IS NOT NULL)
    ->         ),
    ->
    ->     CONSTRAINT chk_maintenance_dates
    ->         CHECK (
    ->             completion_date IS NULL
    ->             OR completion_date >= scheduled_date
    ->         )
    -> );
Query OK, 0 rows affected (0.14 sec)
```

Step 12 – ALTER

SCREENSHOT 13

```
mysql> ALTER TABLE POWER_PLANT
    -> ADD CONSTRAINT uq_plant_name UNIQUE (plant_name);
Query OK, 0 rows affected (0.09 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

Step 13 – CREATE INDEX

SCREENSHOT 14

```
mysql> CREATE INDEX idx_outage_zone
    -> ON OUTAGE(zone_id);
Query OK, 0 rows affected (0.05 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql>
mysql> CREATE INDEX idx_meter_zone
    -> ON METER(zone_id);
Query OK, 0 rows affected (0.04 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql>
mysql> CREATE INDEX idx_allocation_plant
    -> ON ENERGY_ALLOCATION(plant_id);
Query OK, 0 rows affected (0.04 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql>
mysql> CREATE INDEX idx_allocation_zone
    -> ON ENERGY_ALLOCATION(zone_id);
Query OK, 0 rows affected (0.03 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql>
mysql> CREATE INDEX idx_resource_allocation_allocation
    -> ON RESOURCE_ALLOCATION(allocation_id);
Query OK, 0 rows affected (0.03 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql>
mysql> CREATE INDEX idx_log_allocation
    -> ON ALLOCATION_LOG(allocation_id);
Query OK, 0 rows affected (0.03 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql>
mysql> CREATE INDEX idx_log_changed_by
    -> ON ALLOCATION_LOG(changed_by);
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql>
mysql> CREATE INDEX idx_maintenance_performed_by
    -> ON MAINTENANCE(performed_by);
Query OK, 0 rows affected (0.11 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

Step 14 – Verify Indexes

SCREENSHOT 15

```
mysql> SHOW INDEX FROM ENERGY_ALLOCATION;
+-----------------+------------+---------------------+--------------+---------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
| Table           | Non_unique | Key_name            | Seq_in_index | Column_name   | Collation | Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_comment | Visible | Expression |
+-----------------+------------+---------------------+--------------+---------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
| energy_allocation |          0 | PRIMARY             |            1 | allocation_id | A         |           0 |     NULL |   NULL |      | BTREE      |         |               | YES     | NULL       |
| energy_allocation |          1 | idx_allocation_plant |            1 | plant_id      | A         |           0 |     NULL |   NULL |      | BTREE      |         |               | YES     | NULL       |
| energy_allocation |          1 | idx_allocation_zone  |            1 | zone_id       | A         |           0 |     NULL |   NULL |      | BTREE      |         |               | YES     | NULL       |
+-----------------+------------+---------------------+--------------+---------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
3 rows in set (0.05 sec)
```

Step 15 – DROP

SCREENSHOT 16

```
mysql> CREATE INDEX idx_temp_plant_status
    -> ON POWER_PLANT(status);
Query OK, 0 rows affected (0.04 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> DROP INDEX idx_temp_plant_status
    -> ON POWER_PLANT;
Query OK, 0 rows affected (0.09 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

Step 16 – CREATE VIEW

SCREENSHOT 17

```
mysql> CREATE VIEW vw_power_allocation_summary AS
    -> SELECT
    ->     ea.allocation_id,
    ->     pp.plant_name,
    ->     gz.zone_name,
    ->     ea.allocated_power_mw,
    ->     ea.allocation_time,
    ->     ea.priority_level,
    ->     ea.status
    -> FROM ENERGY_ALLOCATION ea
    -> JOIN POWER_PLANT pp
    ->     ON ea.plant_id = pp.plant_id
    -> JOIN GRID_ZONE gz
    ->     ON ea.zone_id = gz.zone_id;
Query OK, 0 rows affected (0.07 sec)

mysql> SELECT * FROM vw_power_allocation_summary;
Empty set (0.00 sec)
```

Step 17 – Final Table Verification

SCREENSHOT 18

```
mysql> SHOW TABLES;
+-----------------------------+
| Tables_in_smart_power_grid  |
+-----------------------------+
| allocation_log              |
| consumer                    |
| energy_allocation           |
| energy_resource             |
| grid_zone                   |
| maintenance                 |
| meter                       |
| outage                      |
| power_plant                 |
| resource_allocation         |
| user                        |
| vw_power_allocation_summary |
+-----------------------------+
12 rows in set (0.01 sec)
```